

BIOLOGY



Biology, Grade 11

SBI3C

This course focuses on the processes that occur in biological systems. Students will learn concepts and theories as they conduct investigations in the areas of cellular biology, microbiology, genetics, the anatomy of mammals, and the structure of plants and their role in the natural environment. Emphasis will be placed on the practical application of concepts, and on the skills needed for further study in various branches of the life sciences and related fields.

Prerequisite: Science, Grade 10, Academic or Applied

Big Ideas

Cellular Biology

- Life processes are determined by the structures and functions of biochemical compounds, cell organelles, and body systems.
- Technological devices that support cellular functions and processes can be used to improve human health.
- Substances that are present in our everyday lives can affect cellular functions and processes in positive and negative ways.

Microbiology

- Groups of microorganisms have common characteristics, and these characteristics enable them to interact with other organisms in the environment
- Microorganisms can have both positive and negative effects on the environment.
- The technological use of microorganisms raises many ethical issues.

Genetics

- Genetic research and biotechnology have social, environmental, and ethical implications.
- Variability and diversity of living organisms result from the distribution of genetic materials during the process of meiosis.

Anatomy of Mammals

- Groups of organs with specific structures and functions work together as systems, which interact with other systems in the body.
- Technologies that are used to maintain human health have social and economic benefits and costs.
- Environmental factors, including natural factors and those resulting from human activity, can have a wide range of effects on human health.

Plants in the Natural Environment

- Plants have specialized structures with distinct functions that enable them to respond and adapt to their environment.
- Plants are critical to the survival of ecosystems.
- Humans affect the sustainability of ecosystems when they alter the balance of plants within those ecosystems.

Fundamental Concepts Covered in This Course (see also page 5)

Fundamental Concepts	Cellular Biology	Microbiology	Genetics	Anatomy of Mammals	Plants in the Natural Environment
Matter	✓				
Energy	✓				
Systems and Interactions		✓		✓	
Structure and Function	✓	✓	✓	✓	✓
Sustainability and Stewardship		✓			✓
Change and Continuity			✓		✓

A. SCIENTIFIC INVESTIGATION SKILLS AND CAREER EXPLORATION

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1.** demonstrate scientific investigation skills (related to both inquiry and research) in the four areas of skills (initiating and planning, performing and recording, analysing and interpreting, and communicating);
- B2.** identify and describe careers related to the fields of science under study, and describe the contributions of scientists, including Canadians, to those fields.

SPECIFIC EXPECTATIONS

A1. Scientific Investigation Skills

Throughout this course, students will:

Initiating and Planning [IP]*

A1.1 formulate relevant scientific questions about observed relationships, ideas, problems, or issues, make informed predictions, and/or formulate educated hypotheses to focus inquiries or research

A1.2 select appropriate instruments (e.g., a microscope, a stethoscope, plant-propagation instruments, dissection instruments) and materials (e.g., prepared slides, agar plates, plants), and identify appropriate methods, techniques, and procedures, for each inquiry

A1.3 identify and locate a variety of print and electronic sources that enable them to address research topics fully and appropriately

A1.4 apply knowledge and understanding of safe laboratory practices and procedures when planning investigations by correctly interpreting Workplace Hazardous Materials Information System (WHMIS) symbols; by using appropriate techniques for handling and storing laboratory equipment and materials and disposing of laboratory and biological materials (e.g., preserved specimens); and by using appropriate personal protection

Performing and Recording [PR]*

A1.5 conduct inquiries, controlling relevant variables, adapting or extending procedures as required, and using appropriate materials and equipment safely, accurately, and effectively, to collect observations and data

A1.6 compile accurate data from laboratory and other sources, and organize and record the data, using appropriate formats, including tables, flow charts, graphs, and/or diagrams

A1.7 select, organize, and record relevant information on research topics from a variety of appropriate sources, including electronic, print, and/or human sources, using suitable formats and an accepted form of academic documentation

Analysing and Interpreting [AI]*

A1.8 synthesize, analyse, interpret, and evaluate qualitative and/or quantitative data to determine whether the evidence supports or refutes the initial prediction or hypothesis and whether it is consistent with scientific theory; identify sources of bias and/or error; and suggest improvements to the inquiry to reduce the likelihood of error

A1.9 analyse the information gathered from research sources for logic, accuracy, reliability, adequacy, and bias

* The abbreviation(s) for the broad area(s) of investigation skills – IP, PR, AI, and/or C – are provided in square brackets at the end of the expectations in strands B–F to which the particular area(s) relate (see pp. 20–22 for information on scientific investigation skills).

A1.10 draw conclusions based on inquiry results and research findings, and justify their conclusions with reference to scientific knowledge

Communicating [C]*

A1.11 communicate ideas, plans, procedures, results, and conclusions orally, in writing, and/or in electronic presentations, using appropriate language and a variety of formats (e.g., data tables, laboratory reports, presentations, debates, simulations, models)

A1.12 use appropriate numeric, symbolic, and graphic modes of representation (e.g., biological diagrams, Punnett squares), and appropriate units of measurement (e.g., SI and imperial units)

A1.13 express the results of any calculations involving data accurately and precisely, to the appropriate number of decimal places and significant figures

A2. Career Exploration

Throughout this course, students will:

A2.1 identify and describe a variety of careers related to the fields of science under study (e.g., food science technologist, medical laboratory technologist, dental hygienist, outpost clinic/primary care nurse, respiratory therapist, veterinary technician, water or wastewater technician) and the education and training necessary for these careers

A2.2 describe the contributions of scientists, including Canadians (e.g., Julia Levy, Charles Beer, Shirley Tilghman, Walter Lewis, Gail Anderson), to the fields under study

B. CELLULAR BIOLOGY

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** evaluate the impact of environmental factors and medical technologies on certain cellular processes that occur in the human body;
- B2.** investigate the structures and functions of cells, and the factors that influence cellular activity, using appropriate laboratory equipment and techniques;
- B3.** demonstrate an understanding of the basic processes of cellular biology.

SPECIFIC EXPECTATIONS

B1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- B1.1** evaluate the effectiveness of medical devices and technologies that are intended to aid cellular functions or processes (e.g., insulin infusion pump, chemotherapy) [AI, C]

Sample issue: For many people with diabetes, insulin pump therapy has replaced multiple daily insulin injections. The pump dispenses accurate doses of insulin and achieves reliable blood sugar control. However, not all jurisdictions fund the pump for all diabetics, and some people prefer self-administered injections to being hooked up to a machine.

Sample questions: How can stem cells be used in the treatment of leukemia? How effective is this type of treatment? Why is laser technology more effective than conventional surgery in removing cancerous tumours? What are some of the effects of kidney dialysis on the renal system and other body systems? What are the advantages of using nanotechnologies in the imaging and diagnosis of cellular abnormalities?

- B1.2** analyse the effects of environmental factors on cellular processes that occur in the human body (e.g., the effect of lead on nerve cells; the effect of electromagnetic radiation on brain cells) [AI, C]

Sample issue: Vitamin D, essential to cellular processes that ensure the health of the bones and teeth, is not well absorbed by the human digestive system. It is manufactured by the body after exposure to the ultraviolet radiation of the sun. However, long-term exposure to the sun without proper UVA and UVB sunscreen protection can eventually lead to skin cancers, such as melanoma.

Sample questions: How might ingesting a high level of mercury by eating contaminated fish affect the nerve cells in our bodies? Which chemicals that are sometimes found in drinking water can affect the cells of the reproductive system? What are their possible effects? How can ultraviolet light from the sun affect the cells of the human eye? What types of toxins accumulate in human cells? What is their long-term effect on the body?

B2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- B2.1** use appropriate terminology related to cellular biology, including, but not limited to: *macro-molecule*, *passive transport*, *active transport*, *catalyst*, and *fluid mosaic model* [C]
- B2.2** investigate the effect of various qualitative factors (e.g., temperature) on the rate of diffusion of molecules across a plasma membrane [PR, AI]

B2.3 using a light microscope, identify visible organelles of a plant cell in a wet mount and an animal cell from a prepared slide, produce an accurate labelled drawing of each cell, and calculate and properly express the magnification of each image [PR, AI, C]

B2.4 investigate the effects of various qualitative factors on the action of enzymes (e.g., the effect of temperature or pH on the breakdown of starch by salivary enzymes) [PR, AI]

B2.5 conduct biological tests to identify biochemical compounds found in various food samples (e.g., use a biuret solution to test for proteins in samples of gelatin and albumin), and compare the biochemical compounds found in each food to those found in the others [PR, AI]

B3. Understanding Basic Concepts

By the end of this course, students will:

B3.1 describe the structures and functions of important biochemical compounds, including carbohydrates, proteins, enzymes, and lipids

B3.2 explain the roles of various organelles, including lysosomes, vacuoles, mitochondria, cell membranes, ribosomes, the endoplasmic reticulum, and Golgi bodies, in the processes of digestion, cellular respiration, and protein synthesis

B3.3 explain the chemical changes and energy transformations associated with the process of cellular respiration, and compare the reactants (i.e., glucose, oxygen) to the products (i.e., water, carbon dioxide, ATP)

B3.4 explain the importance of various cellular processes in human systems (e.g., enzymes act as biological catalysts to regulate chemical processes in the cells of the digestive system)

C. MICROBIOLOGY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** assess the effects of microorganisms in the environment, and analyse ethical issues related to their use in biotechnology;
- C2.** investigate the development and physical characteristics of microorganisms, using appropriate laboratory equipment and techniques;
- C3.** demonstrate an understanding of the diversity of microorganisms and the relationships that exist between them.

SPECIFIC EXPECTATIONS

C1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- C1.1** assess some of the effects, both beneficial and harmful, of microorganisms in the environment (e.g., decomposers break down waste, *E. coli* in water systems poses a severe risk to human health) [AI, C]

Sample issue: Adding beneficial microorganisms to compost at large-scale composting facilities aids in the decomposition of organic waste, and produces high-quality compost in a reduced amount of time. However, the microorganisms can leach into groundwater and run off into nearby water systems, where they can harm other organisms.

Sample questions: In what ways are nitrogen-fixing bacteria beneficial to plants but harmful to other organisms? How can microorganisms that have symbiotic relationships with some organisms be harmful to other organisms?

- C1.2** analyse ethical issues related to the use of microorganisms in biotechnology (e.g., with respect to the use of bacterial insecticides, the patenting of modified microorganisms) [AI, C]

Sample issue: Genetically modified microorganisms are used in many biotechnological applications that benefit humans, in areas such as food production, the development of antibiotics and vaccines, and waste disposal. However, some of the same techniques could be used to manipulate microorganisms to create rapidly reproducing drug-resistant superbugs for use as bioweapons.

Sample questions: Should suppliers be required to label foods that have been modified using microorganisms, so that consumers can make more informed decisions about the food they eat? Why or why not? What do we know about the side effects and possible long-term effects of the medical and cosmetic uses of botulinum toxin? Do you think the use of this substance is adequately regulated?

C2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- C2.1** use appropriate terminology related to microbiology, including, but not limited to: *fission*, *conjugation*, *phage*, *dormancy*, *morphology*, *mycelium*, *spore*, *pathogen*, and *plasmid* [C]
- C2.2** compare and contrast the cell structures of eukaryotes such as fungi, protozoa, and algae [PR, AI]
- C2.3** prepare a laboratory culture of microorganisms (e.g., *acidophilus*) on agar, using proper aseptic techniques [PR]
- C2.4** investigate the effect of antibacterial agents on different bacterial cultures (e.g., the effects of antibacterial soap or mouthwash on a bacterial culture) [PR]
- C2.5** investigate and analyse the conditions (e.g., optimal temperature) needed by microorganisms for growth [PR, AI]

C3. Understanding Basic Concepts

By the end of this course, students will:

- C3.1** describe the anatomy and morphology of various groups of microorganisms (e.g., eukaryotes, prokaryotes, viruses)
- C3.2** explain the differences between the life cycles of eukaryotic and prokaryotic microorganisms in terms of cell division
- C3.3** explain the vital roles of microorganisms in symbiotic relationships with other organisms (e.g., gut bacteria in cows digest cellulose; mycorrhizal fungi penetrate and effectively extend a plant's root system)
- C3.4** explain the different methods of reproduction in various types of bacteria, viruses, and fungi
- C3.5** describe how different viruses, bacteria, and fungi can affect host organisms, and how those effects are normally treated or prevented (e.g., hepatitis viruses can damage the liver, but vaccinations can prevent infections; streptococcus bacteria can cause respiratory infections, which are treated with antibiotics; ringworm is a fungal infection of the skin, treated with fungicides)

D. GENETICS

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** evaluate some social, ethical, and environmental implications of genetic research and related technologies;
- D2.** investigate the process of meiosis, and analyse data related to the laws of heredity;
- D3.** demonstrate an understanding of the process of meiosis, and explain the role of genes in the transmission of hereditary characteristics.

SPECIFIC EXPECTATIONS

D1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- D1.1** evaluate, on the basis of research, some of the social and ethical implications of genetic research and reproductive technologies (e.g., sex selection, harvesting umbilical cord cells) [IP, PR, AI, C]

Sample issue: Scientists are using genetically modified embryonic stem cells to study the role of specific genes in human development and aging and the genetic causes of disease. Some people are concerned that these techniques could one day be used to “design” babies with certain desirable traits, creating a class of genetically enriched individuals and an inequitable society.

Sample questions: Who owns and controls our personal genetic information? Who should have access to our personal genetic information and decide how it will be used? Why is it important to regulate research into, and uses of, genetically based reproductive technologies? What are the ethical implications of using medical data from isolated communities (e.g., First Nations communities, homogeneous island communities) to study the genetic makeup of these communities without their knowledge and/or consent?

- D1.2** evaluate, on the basis of research, some of the effects of genetic research and biotechnology (e.g., genetically modified organisms [GMOs]) on the environment [IP, PR, AI, C]

Sample issue: Farmed salmon can be genetically modified to reach market size in half the time of normal fish, and cost half as much to feed. However, entire populations of wild fish could be endangered by mating with bioengineered fish that are released into the wild, with disastrous consequences for the ecosystem.

Sample questions: What are the risks of growing genetically modified crops near fields where traditional crops are growing? Why have some countries banned genetically modified food crops? What impact has the introduction of herbicide-tolerant plants had on local environments? In what ways can insect-resistant plants both improve agriculture and hurt biodiversity? What are some of the possible effects on the environment of releasing bioengineered insects into the wild?

D2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- D2.1** use appropriate terminology related to genetics, including, but not limited to: *spindle*, *haploid*, *diploid*, *heterozygous*, *homozygous*, *hemophilia*, *gamete*, *ultraviolet radiation*, *carcinogen*, *cancer*, *trisomy*, *somatic cell*, and *zygote* [C]
- D2.2** investigate the process of meiosis, using a microscope or computer simulation, and identify, and draw biological diagrams of, the phases of meiosis [PR, C]

D2.3 solve basic problems in genetics that involve monohybrid crosses, using the Punnett square method [AI, C]

D2.4 compile and analyse qualitative and quantitative data, through laboratory inquiry or computer simulation, on monohybrid crosses, and communicate the results (e.g., record data obtained while performing a “virtual fly” lab, and analyse the results to create a karyotype chart) [PR, AI, C]

D3. Understanding Basic Concepts

By the end of this course, students will:

D3.1 explain the process of meiosis in terms of cell division and the movement of chromosomes

D3.2 explain how the concepts of DNA, genes, chromosomes, alleles, mitosis, and meiosis account for the transmission of hereditary characteristics from generation to generation

D3.3 explain the concepts of genotype, phenotype, dominance, recessiveness, and sex linkage

D3.4 describe some genetic disorders that are caused by chromosomal abnormalities (e.g., non-disjunction) or other genetic mutations

D3.5 describe reproductive technologies such as cloning, artificial insemination, and in vitro fertilization

E. ANATOMY OF MAMMALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** analyse the social or economic impact of a technology used to treat systems in the human body, and the impact of lifestyle choices on human health;
- E2.** investigate, through laboratory inquiry or computer simulation, the anatomy, physiology, and response mechanisms of mammals;
- E3.** demonstrate an understanding of the structure, function, and interactions of the circulatory, digestive, and respiratory systems of mammals.

SPECIFIC EXPECTATIONS

E1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- E1.1** analyse the social or economic impact of a medical device or technology related to the treatment of the human circulatory, respiratory, or digestive system (e.g., a pacemaker, a heart-lung bypass machine, kidney dialysis) [AI, C]

Sample issue: A reliable blood substitute for use during emergencies or blood shortages could save many lives and create huge revenues for its producers. However, clinical trials have shown that patients receiving artificial blood have a significantly higher risk of heart attack than those receiving donated blood, thus increasing medical costs.

Sample questions: Are the costs of the disposable capsules used for capsule endoscopy worth the benefits of detecting gastrointestinal diseases? Why or why not? What social and economic issues arise from the use of respirators to keep gravely injured or terminally ill patients alive?

- E1.2** analyse the impact of various lifestyle choices on human health and body systems (e.g., the impact of excessive alcohol consumption on the liver; of smoking on the respiratory system; of loud noise on the auditory system)

Sample issue: The importance of a sensible diet and regular exercise in maintaining healthy body weight is well known. Yet the popularity of fast food and computer games means that many young people are eating too much fat and are not getting enough physical activity. Consequently, childhood obesity levels are on the rise in Canada.

Sample questions: Does information in the media have an impact on your personal lifestyle choices? Why or why not? Why is it important to limit strenuous outdoor activity on smog days or during heat alerts? What are the dangers of eating undercooked hamburgers?

E2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- E2.1** use appropriate terminology related to animal anatomy, including, but not limited to: *systolic contraction, diastolic pressure, diffusion gradient, inhalation, exhalation, coronary, cardiac, ulcer, asthma, and constipation* [C]
- E2.2** use medical equipment (e.g., a stethoscope, a sphygmomanometer) to monitor a human system, and interpret the data collected [PR, AI]

E2.3 plan and conduct an inquiry to determine the effects of specific variables on the human body (e.g., the effects of exercise and rest on heart rates) [IP, PR, AI]

E2.4 perform a laboratory or computer-simulated dissection of a mammal to identify organs, and explain the relationships between the structures and functions of body systems [PR, AI]

E3. Understanding Basic Concepts

By the end of this course, students will:

E3.1 describe the anatomy and physiology of the circulatory system (including the atrium, ventricles, valves, aorta, pulmonary artery, vena cava, capillaries, veins, arteries, blood cells, and platelets), the mechanisms of blood pressure, and the function of the spleen

E3.2 describe the anatomy and physiology of the respiratory system (including the nasal cavity, trachea, larynx, bronchi, bronchioles, alveoli, and oxygenated and deoxygenated blood) and the mechanisms of gas exchange and respiration

E3.3 describe the anatomy and physiology of the digestive system (including the mouth, epiglottis, esophagus, stomach, intestines, liver, and pancreas), the mechanisms of peristalsis, absorption, and mechanical and chemical digestion, and the function of the kidneys

E3.4 explain some of the mechanisms of interaction between a mammal's different body systems (e.g., the exchange of oxygen and carbon dioxide between the respiratory and circulatory systems)

F. PLANTS IN THE NATURAL ENVIRONMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- F1.** analyse the roles of plants in ecosystems, and assess the impact of human activities on the balance of plants within those ecosystems;
- F2.** investigate some of the factors that affect plant growth;
- F3.** demonstrate an understanding of the structure and physiology of plants and their role in the natural environment.

SPECIFIC EXPECTATIONS

F1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- F1.1** analyse, on the basis of research, and report on ways in which plants can be used to sustain ecosystems [IP, PR, AI, C]

Sample issue: Urban areas place intense pressure on ecosystems. Some of the negative impact can be reduced by cultivating urban forests. These green spaces filter air, water, and sunlight; reduce the “urban heat island” effect; provide habitat for wildlife; and increase biodiversity. However, many cities do not set aside enough land for green spaces because of pressures for urban development.

Sample questions: What are some of the environmental and economic benefits of building urban green spaces on large commercial buildings? What role do native plant species in marshes play in filtering organic waste? How have traditional Aboriginal seed maintenance and distribution practices helped sustain ecosystems in Aboriginal communities?

- F1.2** assess the positive and negative impact of human activities on the natural balance of plants (e.g., crop rotation, the use of fertilizers and herbicides, the introduction of new species) [AI, C]

Sample issue: The greening of cities with a variety of native plant species helps to maintain biodiversity, restore natural landscapes, and

provide food and habitat for local wildlife. However, many urban gardeners introduce non-native plants, which can compete with the native species and may not be hospitable to the local wildlife.

Sample questions: How has increased knowledge about plant growth been applied to improve the resistance of some plants to pests, and allowed those plants to be used in non-native areas? What are the positive and negative effects of such applications? In what ways does monoculture affect the natural balance of plants and the ecosystems they help sustain?

F2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- F2.1** use appropriate terminology related to plants in the environment, including, but not limited to: *xylem*, *phloem*, *chloroplast*, *pistil*, *stamen*, *nitrogen fixation*, and *tropism* [C]
- F2.2** investigate various techniques of plant propagation (e.g., leaf cutting, stem cutting, root cutting, seed germination, traditional Aboriginal practices) [PR]
- F2.3** investigate how chemical compounds (e.g., fertilizers, herbicides, pesticides) and physical factors (e.g., amount of sun and water, quality of soil, pH of soil) affect plant growth [PR, AI]

F2.4 investigate plant tropism by growing and observing plants in a variety of natural and human-made environments [PR]

F3. Understanding Basic Concepts

By the end of this course, students will:

F3.1 describe the structure and physiology of the specialized plant tissues involved in conduction, support, storage, and photosynthesis

F3.2 explain the chemical changes and energy transformations associated with the process of photosynthesis, and compare the reactants (i.e., carbon dioxide, radiant energy, water) to the products (i.e., glucose, oxygen)

F3.3 compare the various means of sexual reproduction (e.g., pollination) and asexual reproduction (e.g., grafting, vegetative propagation, cloning) in plants

F3.4 explain the various roles of plants in the sustainability of the natural environment (e.g., in nutrient cycles, in the water cycle, in erosion control, in wildlife habitats)

F3.5 explain the relationship between the structure of a plant and its external environment, and describe the adaptive attributes that result in natural variation in plant structure (e.g., environmental variables cause variation in leaves within a single plant; in the Arctic, the wild crocus grows close to the ground and is covered with fine hairs)

F3.6 explain the role of plant tropism (e.g., response to stimuli such as light, gravity, and humidity) in a plant's survival